



2.6

Competing Function Model Validation

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Residual sign reveals direction: positive means observed exceeds predicted.

2. Write one precise sentence explaining this principle: Random residual scatter supports a model; curves or trends indicate missed structure.

3. Write one precise sentence explaining this principle: Small errors alone are not enough if assumptions or domain are inappropriate.

4. Write one precise sentence explaining this principle: Consequences of error depend on context, not only magnitude.

Core

5. Observed value is 72 and predicted value is 76. Find the residual.
- _____
- _____
6. Why can RMSE be more sensitive to outliers than MAE?
- _____
- _____
7. A model has tiny in-sample error but implausible long-run behavior. Is it validated?
- _____
- _____
8. State what the formula $e = y - \hat{y}$ tells you and name any required restriction.
- _____
- _____
9. State what the formula $MAE = \frac{1}{n} \sum |e_i|$ tells you and name any required restriction.
- _____
- _____
10. State what the formula $RMSE = \sqrt{\frac{1}{n} \sum e_i^2}$ tells you and name any required restriction.
- _____
- _____

Medium

11. Observed=50, predicted=47. Compute and interpret residual; include relative error when meaningful.
- _____
- _____
- _____
12. Observed=80, predicted=86. Compute and interpret residual; include relative error when meaningful.
- _____
- _____
- _____
13. Observed=12, predicted=10. Compute and interpret residual; include relative error when meaningful.
- _____
- _____
- _____

14. Observed=200, predicted=190. Compute and interpret residual; include relative error when meaningful.

15. Observed=0, predicted=3. Compute and interpret residual; include relative error when meaningful.

Hard

16. Model A residuals: -2,1,0,1,-2. Model B: -4,0,0,0,4. Compare MAE and RMSE qualitatively.

17. Why can a residual plot with all positive values indicate bias?

18. A linear model and exponential model have same MAE. Which may be preferable?

19. Describe the consequence of using an underpredicting flood model.

Difficult

20. Construct residuals with MAE 2 and mean residual 0 using four points.

21. Can a model have zero mean residual but be poor? Give evidence.

22. Model A RMSE=3, model B RMSE=4, but B preserves a physical constraint A violates. Which choose?

Challenge / HOT

23. Prove RMSE is at least MAE for any residual list.

24. Explain why choosing a model on the same data used to fit it can overstate predictive quality.

AP-Style Multiple Choice

25. Residual observed-predicted is negative when model

- (A) underpredicts
- (B) overpredicts
- (C) is exact
- (D) has no domain

26. Random residuals near zero

- (A) support the model
- (B) prove it true
- (C) show causation
- (D) guarantee future accuracy

27. RMSE emphasizes

- (A) small errors only
- (B) large errors
- (C) sign only
- (D) units cancellation

28. MAE has same units as

- (A) input
- (B) output
- (C) squared output
- (D) no units

29. A residual trend indicates

- (A) missed structure
- (B) perfect fit
- (C) randomness
- (D) no data

30. Validation data should ideally be

- (A) the same fitted points only
- (B) separate or held out
- (C) deleted
- (D) all zero

AP-Style Free Response

31. Observed values are 12,18,29,45. Model L predicts 14,21,28,41; model E predicts 11,19,31,44. (a) Compute residuals and MAE. (b) Compare patterns. (c) Recommend.

32. A hospital compares two demand models. A underpredicts by 5 patients on average; B overpredicts by 8. Discuss which metric alone cannot decide and analyze asymmetric consequences.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.